The Hidden Privacy Cost of Valuing Hospitals in Federated Learning: Accounting, a Decision Rule, and Evidence from Real Federations

Ponnaluri Jeyanth, Pudota Jaideep, and M. V. P. Chandra Sekhara Rao

***Abstract*—Hospitals that train a model together need to know what each contributes. Under differential privacy (DP), Shapley values are usually computed by retraining coalitions of hospitals, and the privacy cost of the valuation itself is rarely counted. A hospital whose patients enter r private runs pays for all of them: with ten hospitals a reported per-run $\varepsilon = 1$ costs $\varepsilon \approx 33$ for exact Shapley values and ≈ 34 with 100 sampled permutations, and holding the true budget needs $\sqrt{r}$ times the noise, which destroys the ranking. Valuation must therefore post-process releases whose cost is already counted. We compare such valuations (reconstruction from the DP training run, the basis of GTG-Shapley; a single-model score; and OSA-Shapley, which averages one DP-trained model per hospital) with cheap proxies such as hospital size, on two real federations: 19 prostate-cancer hospitals (5,610 men) and four heart-disease hospitals (920 patients). No method wins everywhere. OSA-Shapley is the most accurate without DP in every setting. Under DP it leads when hospitals hold many patients per model parameter (Spearman $\rho = 0.89$ with the exact ranking at $\varepsilon = 2$, against 0.75 for reconstruction and 0.08 for size); reconstruction is better for a neural network and in the heart-disease federation. Size and summary statistics are unreliable and never detect a hospital with random labels. We derive a decision rule indexed by patients per model parameter and test it in a pre-registered experiment: its direction holds, but OSA-Shapley stays competitive down to fewer patients per parameter than predicted, and we revise the rule.**

***Index Terms*—Federated learning, contribution valuation, Shapley value, differential privacy, privacy accounting, prostate cancer, coronary artery disease.**

I. Introduction

WHETHER a man with a raised PSA needs a prostate biopsy is decided increasingly with risk calculators that combine age, PSA, prostate volume and the MRI PI-RADS score. Such calculators need data from many hospitals, but patient records cannot be pooled across institutions and countries (HIPAA, GDPR). Federated learning (FL) lets hospitals train one model while their records stay local [1], [2]. A recent study on 19 real prostate-cancer cohorts showed that the sustainability of such a federation depends on who participates: large hospitals gain little from joining, and hospitals that do not contribute obtain almost the same model as those that do [3]. A federation that is to last therefore needs a fair way to credit each hospital's contribution, for instance to share costs or rewards.

The Shapley value is the standard answer: it is the only allocation that is efficient, symmetric and gives nothing to a hospital that never helps [4], [5]. Hospitals also need differential privacy (DP), because shared model updates can reveal patients [6], [7]. The two requirements interact badly. Exact Shapley values need a model for every coalition of hospitals, and the usual way to study valuation under DP retrains each coalition with DP-SGD and reports a per-run privacy budget. But every hospital then contributes its patients to 2^{K-1} private training runs, so by composition its real privacy loss is far larger than the reported one. Holding the real loss at the stated budget requires so much noise that the ranking is lost. A practical objection is that hospitals could simply share their sizes or a few summary statistics instead of running any valuation protocol; whether such proxies suffice is an empirical question that we answer. Two lines of work point to a way out: reconstructing coalition models from a single training run [8]–[10], and re-using one private release so that noise is shared across estimates [11]. Neither has been evaluated on real hospitals at an honestly accounted budget.

We ask two questions: what does valuation under DP really cost, and which valuation should a federation of real hospitals use at an honest budget? Our contributions are:

- **The hidden cost, made explicit.** Proposition 2 gives the composed privacy cost of any retraining-based valuation in which a hospital's patients enter r private runs, and the $\sqrt{r}$-fold noise needed to keep it honest. For exact Shapley values $r = 2^{K-1}$; Monte-Carlo sampling reduces r only for large K and never removes the problem (Table II). A per-run $\varepsilon = 1$ costs about 33 for exact Shapley values over ten hospitals, and 52 for 19 hospitals even with 100 sampled permutations.
- **A decision rule from real federations.** We compare every valuation that costs the stated budget once, and cheap proxies that cost nothing, on two real federations without simulated or partitioned silos: the 19 prostate-cancer hospitals of a published hospital-participation study and the four hospitals of the UCI Heart Disease collection. Which valuation is most accurate depends on one quantity, the number of training patients per hospital relative to the square root of the number of model parameters, which Proposition 3 identifies as the signal-to-noise ratio of a DP release. We wrote down where the rule's switch-over should lie before testing it

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Code, data provenance and all results: https://github.com/JeyanthPonnaluri/Mini_Project_RVR.

on new conditions; four of six pre-registered hypotheses held, the predicted threshold did not, and the revised rule (Section VII) tells a federation when to use OSA-Shapley, when reconstruction, when size suffices, and when no privacy-honest valuation is reliable.

- **OSA-Shapley.** One of the options is new: each hospital releases one DP-trained model and every coalition model is the size-weighted average of these releases, so exact valuation of all 2^K coalitions costs the budget once (Proposition 1). It is the most accurate valuation without DP in every setting we studied and, for the ten largest prostate-cancer hospitals, also under DP ($\rho = 0.89$ at $\varepsilon = 2$; OSA$-$MR $= +0.14$ $[+0.08, +0.19]$); it never fails to flag a hospital with random labels at $\varepsilon \geq 2$.
- **Honest negative results.** Hospital size, case counts and summary statistics do not identify contributions reliably (ρ from -0.38 to 0.74) and never detect bad data. With few patients per parameter, OSA-Shapley loses to reconstruction (heart disease: ρ 0.15 vs 0.45 at $\varepsilon = 2$; federated MLP: 0.65 vs 0.82), and at $\varepsilon = 1$ no valuation recovers the heart-disease ranking.

Section II reviews related work, Sections III–IV define the setting and the valuation methods, Section V describes the data and protocol, Section VI reports the results, Section VII derives and tests the decision rule, and Sections VIII–IX discuss them.

II. Related Work

A. Federated Learning across Hospitals

Cross-silo FL trains clinical models across institutions without sharing records [1], [12]–[14]; hospitals differ in case mix, so local models drift from the global one [15]–[17], which FedProx, SCAFFOLD and FedAdam address [18]–[21]. Real-silo benchmarks such as FLamby [22] keep the natural hospital split. Kazlouski *et al.* [3] studied, on 19 real prostate-cancer cohorts, when hospitals benefit from joining a federation and how free-riders fare, which is the setting we value.

B. Differential Privacy and Accounting

DP bounds the influence of any single record [23], [24]. DP-SGD clips per-record gradients and adds Gaussian noise [7]; Rényi DP composes the loss tightly over many steps [25], and Poisson subsampling amplifies privacy [26], [27]. Cross-silo variants protect users across hospitals [28]. Membership-inference attacks provide an empirical audit [6], [29], [30].

C. Contribution Valuation in Federated Learning

Leave-one-out compares the federation with and without a client and gives near-zero value to redundant clients [5]. The Shapley value [4] averages marginal contributions over all coalitions; it has been adapted to data valuation [5], [31], [32] and to federated clients [8], [9], [33]. Because it needs a model per coalition, federated valuation either samples coalitions [34], [35] or reconstructs coalition models from the updates of one training run: the multi-round (MR) and one-round (OR) reconstructions of Song *et al.* [8] and their truncated Monte-Carlo approximation GTG-Shapley [10]. Zhou *et al.* [11] show that independent DP noise makes data values unreliable as the estimation budget grows, and that re-using already released private gradients avoids this at no extra privacy cost. Threshold KNN-Shapley offers native DP for nearest-neighbour models [36]. Studies of valuation under DP usually retrain coalitions at a per-run budget. Reconstruction from a DP training run is post-processing and therefore privacy-free, but, to our knowledge, it has not been compared with retraining and with other post-processing valuations at the same true budget on real hospitals (Table I).

TABLE I
POSITION OF THIS WORK. *Cost counted*: THE PRIVACY COST OF THE VALUATION ITSELF IS ACCOUNTED.

Study	Real hospitals	DP	Valuation	Cost counted	vs reconstruction	vs cheap proxies
McMahan *et al.* [2]	No	No	No	–	–	No
Abadi *et al.* [7]	No	Yes	No	–	–	No
Ghorbani & Zou [5]	No	No	Yes	–	No	No
Song *et al.* [8]	No	No	Yes	–	–	No
Liu *et al.* (GTG) [10]	No	No	Yes	–	–	No
Zhou *et al.* [11]	No	Yes	Yes	Yes	No	No
Kazlouski *et al.* [3]	Yes	No	No	–	–	No
This work	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

III. System Model

A federation has K hospitals and a coordinating server. Hospital k holds $D_k = \{(x_{i,k}, y_{i,k})\}_{i=1}^{n_k}$, where $x_{i,k} \in \mathbb{R}^d$ are pre-biopsy variables and $y_{i,k} \in \{0, 1\}$ indicates the outcome (clinically significant prostate cancer, or coronary artery disease in the second federation); $n = \sum_k n_k$. The federation minimises

$$\min_w F(w) = \sum_{k=1}^{K} \frac{n_k}{n} F_k(w), \tag{1}$$

with $F_k(w) = \frac{1}{n_k} \sum_{i=1}^{n_k} \ell(w; x_{i,k}, y_{i,k}) + \frac{\lambda}{2}\|w\|_2^2$. Under FedAvg each hospital runs E local epochs from the current global model and the server averages, $w^{t+1} = \sum_k (n_k/n)\, w_k^{t+1}$; FedProx adds $\frac{\mu}{2}\|w - w^t\|_2^2$ to the local objective. Local gradients are clipped and perturbed with Gaussian noise calibrated by a Rényi-DP accountant. The server then values each hospital with the Shapley value of a utility $v(S)$, the AUC on a reference set of a model representing coalition S. How that coalition model is obtained, and what it costs in privacy, is the subject of this paper (Fig. 1).

IV. Methods

A. Classification Model

Risk is modelled with logistic regression with an intercept,

$$p_i = \sigma(z_i) = \frac{1}{1 + e^{-z_i}}, \qquad z_i = x_i^T w + b, \tag{2}$$

trained on binary cross-entropy $\ell(w; x_i, y_i) = -[y_i \ln p_i + (1 - y_i) \ln(1 - p_i)]$ with a small L_2 penalty ($\lambda = 10^{-3}$).

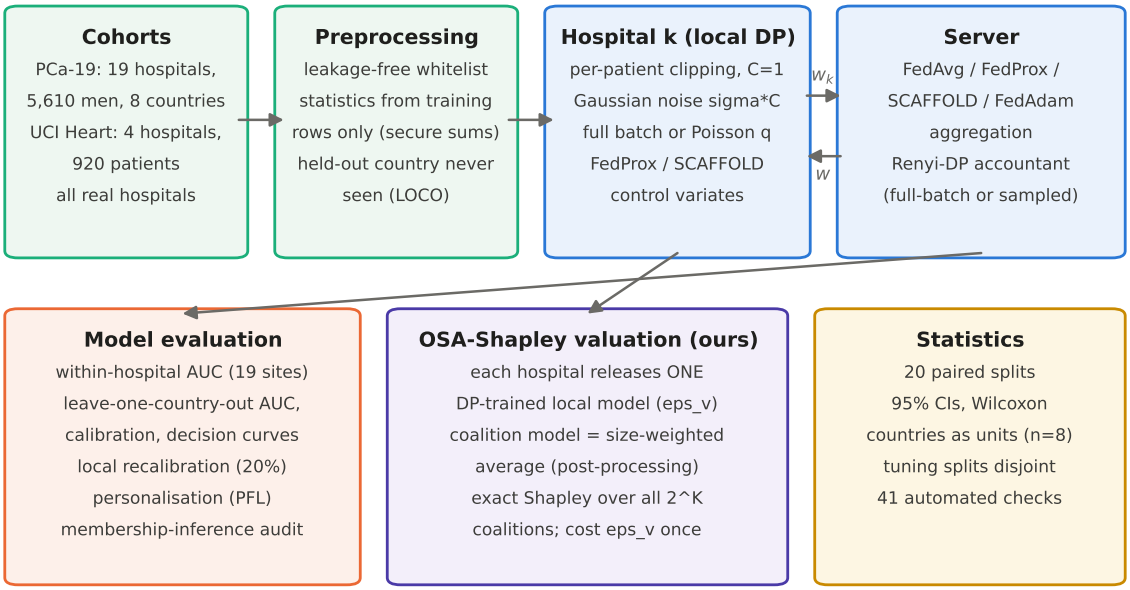


Fig. 1. System architecture. Hospitals train locally with per-patient clipping and Gaussian noise and the server aggregates under a Rényi-DP accountant. The federated model is evaluated within hospitals and on held-out countries (Supplementary Material). Hospitals are valued with OSA-Shapley, which post-processes one DP-trained model per hospital, and compared with retraining and with reconstruction from the training run.

B. Rényi Differential Privacy and Sensitivity Analysis

1) Privacy Unit and Adjacency: We use add/remove adjacency of single records, $D \sim D' \iff |D \triangle D'| = 1$. Every patient contributes one record, so the guarantee is patient-level.

Each per-sample gradient is clipped to L_2 norm C,

$$\bar{g}_i = g_i / \max\left(1, \frac{\|g_i\|_2}{C}\right), \tag{3}$$

so adding or removing one patient changes the summed gradient by at most C. The released (noisy) gradient is

$$\tilde{g} = \frac{1}{n_k}\Big(\sum_{i=1}^{n_k} \bar{g}_i + z\Big), \quad z \sim \mathcal{N}\left(0, \sigma^2 C^2 I\right), \tag{4}$$

i.e. a Gaussian mechanism with noise multiplier σ on a sum query of sensitivity C (client sizes n_k are treated as public). The L_2 and proximal terms do not access data and are post-processing.

2) Rényi DP Composition: Updates are full-batch (sampling rate $q = 1$), so no subsampling amplification is claimed. A Gaussian mechanism with multiplier σ satisfies RDP of order $\alpha > 1$ with $\epsilon_{RDP}(\alpha) = \alpha/(2\sigma^2)$. Every patient participates in each of the $N = T \times E$ local steps of its own client, and clients hold disjoint patients, so

$$\epsilon = \min_{\alpha \in \mathcal{A}} \left\{ N \cdot \frac{\alpha}{2\sigma^2} + \frac{\ln(1/\delta)}{\alpha - 1} \right\}, \tag{5}$$

with a dense order grid $\mathcal{A}$ (1.1–4096). The accountant agrees with the closed-form optimum $\rho + 2\sqrt{\rho \ln(1/\delta)}$, $\rho = N/(2\sigma^2)$, to within 0.01% and is never below it. For $N = 100$ and $\delta = 10^{-5}$ the calibrated multipliers are $\sigma = 5.68, 10.55, 24.99, 49.02, 97.01$ for $\varepsilon = 10, 5, 2, 1, 0.5$.

TABLE II
THE PRIVACY COST THAT RETRAINING-BASED VALUATION ACTUALLY SPENDS (PROPOSITION 2). r: NUMBER OF DP TRAINING RUNS THAT CONTAIN A GIVEN HOSPITAL'S PATIENTS (EXACT SHAPLEY: 2^{K-1}; PERMUTATION SAMPLING WITHOUT CACHING: ON AVERAGE $100(K+1)/2$). EACH RUN IS $N = 100$ FULL-BATCH DP-GD STEPS CALIBRATED TO THE PER-RUN ε; THE TABLE GIVES THE COMPOSED ε PER HOSPITAL ($\delta = 10^{-5}$). LAST COLUMN: FACTOR BY WHICH EVERY RUN'S NOISE MUST GROW TO KEEP THE STATED BUDGET ($\sqrt{r}$). POST-PROCESSING VALUATIONS SPEND THE STATED BUDGET ONCE.

		True ε for a per-run ε of				Noise
Valuation	r	10	5	2	1	factor
Exact, $K = 4$	8	36	16	6.1	2.9	2.8
Exact, $K = 10$	512	988	334	85	33	23
Exact, $K = 19$	$\approx 10^5$	$\approx 10^5$	$\approx 10^5$	23,197	6,116	512
Exact, $K = 50$	$\approx 10^{14}$	$\approx 10^{14}$	$\approx 10^{14}$	$\approx 10^{13}$	$\approx 10^{13}$	$\approx 10^7$
100 permutations, $K = 10$	550	1,053	354	89	34	23
100 permutations, $K = 19$	1,000	1,820	597	141	52	32
100 permutations, $K = 50$	2,550	4,464	1,376	303	103	50
OSA-Shapley, MR, OR (any K)	1	10	5	2	1	1

Poisson-subsampled DP-SGD with sampled-Gaussian accounting is evaluated as a sensitivity analysis in the Supplementary Material.

C. Exact Shapley Valuation

The Shapley value of client k is

$$\phi_k = \sum_{S \subseteq \mathcal{N} \setminus \{k\}} \frac{|S|!(K - |S| - 1)!}{K!} \left[v(S \cup \{k\}) - v(S) \right], \tag{6}$$

with $v(\emptyset) = 0.5$. We compute it *exactly* by training all $2^K - 1$ coalitions (vectorised across coalitions; numerically identical to separate training runs) for $K \leq 10$, so no Monte-Carlo error confounds the effect of DP noise. Leave-one-out (LOO)

is $v(\mathcal{N}) - v(\mathcal{N} \setminus \{k\})$. Under DP every coalition run draws independent noise. Note that this *retraining* approach releases 2^{K-1} privately trained models that involve each hospital; by composition, its total privacy loss is far larger than the per-run ε (for $K = 10$, $N = 100$ and a per-run $\varepsilon = 1$ it is $\varepsilon \approx 33$; Table S10). We report it because it is the standard way to study valuation under DP, and contrast it with the privacy-honest estimator below.

D. Privacy-Honest Valuation: OSA-Shapley

We propose to value hospitals with *one-shot-averaging Shapley* (OSA-Shapley), which spends the privacy budget once:

1) Each hospital k releases one logistic-regression model $\hat{w}_k$ trained locally by DP gradient descent at valuation budget ε_v (30 full-batch steps, per-patient clipping $C = 1$, Gaussian noise calibrated with the accountant above). The released model is the average of the iterates of the second half of training, which damps the injected noise.
2) The server forms the model of every coalition as the size-weighted average $w_S = \sum_{k \in S} (n_k / n_S)\, \hat{w}_k$. For a convex model trained to convergence this one-shot average approximates the FedAvg solution of the coalition; the base paper itself trains its federated models this way.
3) $v(S)$ is the AUC of w_S on the server's reference set, and exact Shapley values follow from all $2^K - 1$ utilities by vectorised bitmask summation.

Assumption. As for retraining-based valuation, the reference set on which $v(S)$ is computed is held by the server (for example a curated or consented validation set); the guarantees below protect the hospitals' training patients.

Proposition 1 (privacy of OSA-Shapley). If each released model $\hat{w}_k$ is (ε_v, δ)-DP with respect to the patients of hospital k, then the utilities of all $2^K - 1$ coalitions and all Shapley values are jointly (ε_v, δ)-DP with respect to every training patient, for any K.

Proof. Hospitals hold disjoint patients, so a single patient affects only one released model; all utilities and Shapley values are deterministic functions of $(\hat{w}_1, \ldots, \hat{w}_K)$ and the server's reference set. The claim follows from the post-processing property of DP [24]. □

Proposition 2 (cost of retraining-based valuation). Suppose a valuation trains r_k DP models that each use the patients of hospital k, every run consisting of N steps of a full-batch Gaussian mechanism with noise multiplier σ (per-run guarantee (ε, δ)). Then hospital k's total RDP is that of a single run with multiplier $\sigma/\sqrt{r_k}$; equivalently, keeping the total guarantee at (ε, δ) requires every run to use multiplier $\sqrt{r_k}\,\sigma$. For exact Shapley values $r_k = 2^{K-1}$ (a factor of 22.6 for $K = 10$); for Monte-Carlo permutation sampling with M permutations and no caching, r_k is on average $M(K+1)/2$.

Proof. The Gaussian mechanism has RDP $\alpha/(2\sigma^2)$ per step and RDP composes additively, so $r_k N$ steps give $r_k N \alpha/(2\sigma^2) = N\alpha/(2(\sigma/\sqrt{r_k})^2)$; the conversion to (ε, δ) depends on the composed RDP only. In a random permutation hospital k is equally likely to be at any of the K positions and joins every prefix from its position on, i.e. $(K+1)/2$ of the K prefixes on average. □

Table II evaluates Proposition 2. With ten hospitals a per-run $\varepsilon = 1$ costs each hospital $\varepsilon \approx 33$; with 19 hospitals exact retraining is out of reach, and even sampling 100 permutations costs $\varepsilon \approx 52$. The calibrated multipliers in our experiments follow this ratio exactly (e.g. $\sigma = 5.68$ per run versus 128.5 budget-matched at $\varepsilon = 10$, $K = 10$). If the valuation is released together with the training run, the two budgets compose; OSA-Shapley adds ε_v, retraining adds the much larger composed cost of Proposition 2.

Approximation. OSA-Shapley replaces the FedAvg model of each coalition by the average of locally trained models. For i.i.d. data and smooth, strongly convex losses such as regularised logistic regression, this average has mean-squared error $O(n^{-1} + (n/m)^{-2})$ for n samples on m sites, matching centralised training when $m \le \sqrt{n}$ [37]. Hospitals are not i.i.d., so we measure the approximation directly: without DP, OSA-Shapley reproduces the retraining Shapley ranking with Spearman $\rho = 0.96$ for the ten largest and 0.95 for the ten smallest PCa-19 hospitals (Sections VI-C and VI-H). *Noise.* Every coalition shares the same released models, so the injected noise is common to all coalitions and largely cancels in marginal contributions. This is the mechanism that Zhou *et al.* [11] exploit by re-using released private gradients, here realised by re-using one released model per hospital in every coalition. *Cost.* Only K local trainings are needed, and the 2^K evaluations are cheap: exact Shapley values for all 19 PCa-19 hospitals (524,287 coalitions) take about a minute on one CPU core. The number of local steps was chosen on four tuning splits that are disjoint from the evaluation splits.

Proposition 3 (noise of coalition models). Let each release be $\hat{w}_k = w_k + \xi_k$, where the DP perturbations ξ_k are independent, zero-mean, and, for N steps of DP gradient descent with step size η, have per-coordinate standard deviation at most $s_k = \eta\sigma C\sqrt{N}/n_k$ (noise $\mathcal{N}(0, \sigma^2 C^2)$ is added to a gradient sum that is divided by n_k; contraction of the loss and iterate averaging only reduce it). Then the perturbation of every coalition model, $\hat{w}_S - w_S = \sum_{k \in S} (n_k/n_S)\xi_k$, has per-coordinate standard deviation

$$\mathrm{sd}(\hat{w}_S - w_S) = \frac{1}{n_S}\Big(\sum_{k \in S} n_k^2 s_k^2\Big)^{1/2} \le \frac{\eta\sigma C\sqrt{|S|\,N}}{n_S}. \quad (7)$$

Proof. The ξ_k are independent, so the variance of the weighted sum is $\sum_{k \in S} (n_k/n_S)^2 s_k^2$; substituting the bound on s_k gives $\eta^2\sigma^2 C^2 N|S|/n_S^2$. □

Proposition 3 makes two points. First, the noise in a coalition model does not depend on K, whereas budget-matched retraining multiplies it by $2^{(K-1)/2}$ (Proposition 2). Second, it shrinks as coalitions grow (for m hospitals of equal size n the bound is $\eta\sigma C\sqrt{N}/(n\sqrt{m})$), and it is largest for a single small hospital. OSA-Shapley is therefore accurate when hospitals have hundreds of patients and loses its advantage when they have a few dozen, which is what we observe on the smallest real hospitals (Section VI-H). A validation check confirms the bound empirically (the observed standard deviation is 0.39–0.49 of it).

Competitors at the same privacy cost. OSA-Shapley is not the only valuation that is pure post-processing. In DP-FedAvg every hospital releases a noisy local model in every round, and the accountant already covers all $T \cdot E$ steps. Coalition models can therefore be reconstructed from one private training run at no extra cost [8]–[10]. We implement the two reconstructions of Song *et al.* [8] exactly. ***MR*** (multi-round) computes, in every round t, exact Shapley values of $v_t(S) = \mathrm{AUC}\big(\sum_{k\in S}(n_k/n_S)\hat{w}_k^t\big)$ with $v_t(\emptyset)$ the AUC of the global model before the round, and sums them over rounds. GTG-Shapley [10] is a truncated Monte-Carlo approximation of MR, so we evaluate its idealised exact version, MR. ***OR*** (one-round) rebuilds $w_S = w^0 + \sum_t \sum_{k\in S}(n_k/n_S)(\hat{w}_k^t - w^{t-1})$ from the accumulated updates. Both use the same DP-FedAvg run, at the same ε, as the federated model.

Cheap proxies. To test whether a valuation protocol is needed at all, we compare OSA-Shapley with scores that hospitals could share more cheaply: (i) hospital size n_k; (ii) the number of cancer cases; (iii) the similarity of summary statistics, the negative distance between a hospital's feature means and prevalence and the pooled ones; (iv) a self-reported AUC of the hospital's own model on its own test patients; (v) the AUC of its single released DP model on the reference set; and (vi) the leave-one-out value of the averaged releases, $v(\mathcal{N}) - v(\mathcal{N}\backslash\{k\})$. Proxies (v) and (vi) use exactly the releases of OSA-Shapley and therefore cost the same ε; proxies (i)–(iii) are shared without DP and (iv) cannot be verified by the server.

Accuracy. We compare each valuation's ranking with the exact non-private retraining Shapley ranking by Spearman's ρ and report how often a hospital with random labels is ranked last.

E. Statistical Analysis

Every experiment is repeated over 20 seeds. A seed re-draws the stratified 80/20 split within every hospital, model noise and DP noise, so seed-to-seed variation covers all sources of randomness. We report means with t-based 95% confidence intervals over seeds. Comparisons between methods are *paired* by seed; we report the mean paired difference with its 95% CI and a two-sided Wilcoxon signed-rank test, and call a difference significant only if its CI excludes zero.

V. Data and Protocol

A. PCa-19: The Base Paper's 19 Hospitals

PCa-19 is the patient-level collection used by the base paper, released in the authors' public repository (MIT licence) [3]. The authors harmonised it from the supplementary files of 19 peer-reviewed prostate-cancer studies. It contains 5,610 men from 19 hospitals (Table S4); 2,146 (38.3%) have clinically significant cancer (csPCa, ISUP grade group ≥ 2 at biopsy), ranging from 12% to 60% per hospital. The base paper's text states nine countries, but its Table I and the released data contain eight (Spain, South Korea, Germany, China, USA, Netherlands, Italy, UK); we use the released data unchanged. Every variable is available before biopsy: age, PSA, prostate volume (PV) and the highest PI-RADS score. (A 5-α-reductase-inhibitor flag is 0 for all patients and is not used.) Our feature set is age, $\log$ PSA, $\log$ PV and indicators for PI-RADS 3, 4 and 5. The logarithms keep the extreme PSA values of one hospital (median 277 ng/mL) from dominating, and standardisation statistics are computed from training rows only. For the reproduction of the authors' protocol we use their exact Ettala and Noh variables.

B. Second Federation: UCI Heart Disease

The UCI Heart Disease collection [38], [39] contains 920 patients referred for coronary angiography at four hospitals: the Cleveland Clinic Foundation (USA, 303), the Hungarian Institute of Cardiology in Budapest (294), the University Hospitals of Zurich and Basel (123) and the VA Medical Center in Long Beach (USA, 200). It is a standard real-silo benchmark for federated learning [22]. The target is angiographic coronary artery disease (any vessel narrowed by more than 50%), whose prevalence ranges from 36% (Budapest) to 93% (Switzerland). We drop the three variables with more than 30% missing values (slope, number of vessels, thallium test), treat a cholesterol of zero as missing (all Swiss patients), impute remaining gaps with the median of the pooled training rows plus a missingness indicator, one-hot encode chest-pain type and resting ECG, and standardise with training-row statistics, giving 20 features. OSA-Shapley keeps the hyper-parameters chosen on PCa-19, so this cohort is an out-of-sample test of the method as well as of the decision rule.

C. Silos, Splits and Reference Sets

Every client is a real hospital exactly as released (one of the 19 PCa-19 hospitals or the four heart-disease hospitals); we never simulate hospitals or partition a pooled data set, so every valued unit is an institution that could join a federation. Within each hospital patients are split 80/20 into training and test sets, stratified by outcome, and a new split is drawn for every seed. We value two groups for which the exact non-private reference is computable (all $2^{10} - 1$ coalitions retrained): the ten largest hospitals (266–763 patients; the Dutch hospital, the tenth largest, also belongs to the smallest ten) and the ten smallest (16–266 patients; nl, it, es4, es5, de2, gb, es1, es8, es3, es2). All 19 hospitals are valued with OSA-Shapley only, because no exact reference exists for them. In the heart-disease federation all four hospitals are valued against the exact reference ($2^4 - 1$ coalitions). The reference set on which $v(S)$ is scored is the pooled test patients of the valued hospitals and is never used for training. The only controlled perturbation is the "useless hospital" of Section VI-F, a real hospital whose training labels are replaced by random labels at its own prevalence (es6 in PCa-19, Budapest in the heart-disease federation).

D. Protocol

Unless stated otherwise: $T = 20$ rounds, $E = 5$ local epochs ($N = 100$ steps), learning rate $\eta = 0.5$, $\lambda = 10^{-3}$, $\mu = 0.5$, $C = 1$, $\delta = 10^{-5}$, 20 seeds. These optimiser settings were fixed before the valuation experiments and are shared

by every method; OSA-Shapley's single hyper-parameter (the number of local steps) was chosen on four tuning splits.

Valuation experiments use 20 further splits (seeds 20–39) that were not used for any tuning.

VI. Results

All results use real hospitals as released: the 19 PCa-19 hospitals and, in Section VI-I, the four heart-disease hospitals. We first summarise the federation that is valued (Section VI-A); its full analysis is in Supplementary Section S2. We then turn to valuation under DP (Sections VI-B–VI-I).

A. The Federation Being Valued

Our re-implementation of the base paper's protocol reproduces the authors' released code to within 1.2×10^{-14} for both of their calculators (Supplementary Section S2-A), and its sustainability finding holds under our protocol: joining brings no significant average gain over local training (FL$-$LOC $= -0.005$ $[-0.015, +0.006]$), a free-rider obtains almost the same model (FR$-$FL $= -0.002$ $[-0.002, -0.001]$), and federated training matches pooled training (CEN$-$FL $= +0.002$ $[-0.001, +0.004]$). For this seven-parameter, regularised logistic calculator trained on 5,610 patients, DP changes the mean per-hospital AUC by only -0.005 $[-0.011, +0.001]$ even at $\varepsilon = 0.5$. We do not generalise this: the cost of a given ε depends strongly on the model and the amount of data, and for a federated MLP the same budgets perturb the released models far more (Section VI-G). Discrimination transports to held-out countries, but calibration does not (observed/expected ratios 0.47–1.30); an intercept update on 20% of a new country's patients, which is common clinical practice for risk calculators anyway, reduces $|\log \mathrm{O/E}|$ from 0.263 to 0.071 (Supplementary Section S2-C).

B. Retraining-Based Valuation under DP

Following standard practice, we first retrain every coalition with DP at a per-run ε and compute exact Shapley values over the ten largest hospitals (1,023 coalitions). Their agreement with the non-private ranking falls from Spearman $\rho = 0.94$ at $\varepsilon = 10$ to 0.91 at $\varepsilon = 2$ and 0.64 at $\varepsilon = 0.5$ (original splits, against the no-DP ranking; Table S10 repeats this on the held-out splits against exact Shapley), although the federation's AUC changes by less than 0.01 over this range. Leave-one-out values are near zero for every hospital and cannot separate contributors (Spearman with Shapley 0.66 [0.56, 0.77]). These results ignore the cost of the valuation itself.

C. Valuation at an Honest Privacy Budget

Table S10 makes the real cost of retraining explicit: with ten hospitals each hospital joins 512 private training runs, so a per-run ε of 10, 5 or 1 corresponds to a total ε of about 988, 334 and 33 per hospital. When the total budget is held at the stated ε (*budget-matched* retraining), every run needs much more noise and the ranking is lost: Spearman ρ with the exact non-private Shapley values falls to 0.47 at $\varepsilon = 10$ and 0.10 at $\varepsilon = 1$.

OSA-Shapley (Section IV-D) spends the budget once. Without DP it reproduces the retraining Shapley ranking almost exactly ($\rho = 0.96$), and under DP it keeps $\rho = 0.90$ at $\varepsilon = 10$, 0.89 at $\varepsilon = 2$, 0.80 at $\varepsilon = 1$ and 0.54 at $\varepsilon = 0.5$ (Table S10; evaluated on 20 splits that were not used to choose its single hyper-parameter). At equal privacy it beats retraining by $+0.71$ $[+0.58, +0.84]$ in ρ at $\varepsilon = 2$. At $\varepsilon = 2$ it matches per-run retraining (-0.02 $[-0.06, +0.03]$), which spends 28–42 times more privacy. Because no coalition needs training, exact valuation of all 19 hospitals is feasible: the OSA ranking computed with DP agrees with the non-private one at $\rho = 0.95$ ($\varepsilon = 5$) and 0.80 ($\varepsilon = 1$).

D. Against Valuations that Also Cost the Budget Once

The fair competitors are the reconstruction methods MR and OR, which post-process the DP-FedAvg run (Section IV-D). Table S12 and Fig. S5 compare them on the same splits. For the ten largest hospitals, OSA-Shapley is more accurate than exact MR, the upper bound for GTG-Shapley, at every privacy level: ρ 0.89 vs 0.75 at $\varepsilon = 2$ (difference +0.14 [+0.08, +0.19]) and 0.80 vs 0.65 at $\varepsilon = 1$ (+0.15 [+0.03, +0.26]). It also beats OR by +0.37 [+0.26, +0.48] at $\varepsilon = 2$. The gap is already present without DP (ρ 0.96 vs 0.74), so it comes from the coalition proxy rather than from the noise.

E. Is OSA-Shapley Worth It? Cheap Proxies

A federation could avoid a valuation protocol altogether and credit hospitals by what they can share cheaply. Table III compares these proxies with OSA-Shapley against the same exact non-private reference. For the ten largest hospitals, hospital size, the number of cancer cases and the similarity of summary statistics to the pooled ones carry almost no information about contribution (ρ 0.08, 0.02 and -0.18; OSA$-$size $= +0.80$ $[+0.70, +0.91]$ at $\varepsilon = 2$). They also cannot detect a hospital with useless data, because random labels change neither its size nor its prevalence (0% of splits). Hospitals here differ in case mix, not only in size, and their marginal value depends on what the other hospitals already provide. An AUC that each hospital reports for its own model is unverifiable and agrees poorly with the ranking ($\rho = 0.25$ at $\varepsilon = 2$).

Proxies built on a released model behave differently. Once each hospital has released one DP model, as OSA requires, the AUC of that single model on the reference set already ranks the large hospitals well ($\rho = 0.85$ at $\varepsilon = 2$) and flags the useless hospital. OSA-Shapley is comparable at $\varepsilon = 2$ ($+0.04$ $[-0.00, +0.08]$) and better at $\varepsilon = 1$ ($+0.10$ $[+0.05, +0.14]$), where single releases are noisy and averaging over coalitions helps; it also beats the leave-one-out value of the averaged releases by $+0.11$ $[+0.08, +0.15]$. Both scores use the same releases, so OSA's advantage costs no extra privacy or communication. OSA's values are also Shapley values: they sum to the federation's gain over chance, which a single-model score does not.

On the ten smallest hospitals the picture changes (Table III, right). There contribution is driven mainly by data volume, and hospital size alone reaches $\rho = 0.74$. At $\varepsilon = 2$, OSA-Shapley, MR, the single-model score and size agree with the ranking

about equally well (OSA$-$size $= +0.05$ $[-0.01, +0.10]$, OSA$-$MR $= +0.00$ $[-0.06, +0.06]$), and at $\varepsilon = 1$ none of the DP valuations is better than size.

F. A Hospital with Useless Data, Removal and Robustness

We replaced the training labels of one mid-sized hospital (es6) by random labels and asked whether each valuation ranks it last (Tables III and S9). The non-private retraining Shapley values always do (100%). At $\varepsilon = 2$ and $\varepsilon = 1$, OSA-Shapley ranks it last in 100% and 95% of splits, MR in 70% and 80%, and budget-matched retraining in only 60% and 45%; no size- or statistics-based proxy ever does. Sensitivity to the reference-set size and the number of local steps is reported in Table S9.

G. A Non-Convex Model

OSA-Shapley assumes that averaging locally trained models approximates the federated model of a coalition, which holds for convex losses. We therefore repeated the comparison with a one-hidden-layer MLP (16 tanh units) trained by FedAvg, with exact non-private retraining Shapley values of the federated MLP as the reference (Table S11). Because averaging the parameters of separately trained networks is fragile, the coalition prediction is the size-weighted average of the members' log-odds (OSA-ens.), which is equally post-processing. Without DP the ensemble variant reproduces the reference ranking well ($\rho = 0.91$, against 0.78 for MR). Under DP it loses this advantage: at $\varepsilon = 2$ and $\varepsilon = 1$ it reaches 0.65 and 0.60, whereas MR reaches 0.82 and 0.78 (OSA$-$MR $= -0.17$ $[-0.25, -0.09]$ at $\varepsilon = 2$). A non-linear model trained alone for 200 noisy steps accumulates the DP perturbation, whereas MR's short local updates restart from the shared global model in every round. Budget-matched retraining of the MLP is close to chance ($\rho = 0.16$; 5 splits). For non-convex models, reconstruction from the training run is the better privacy-honest valuation.

H. Small Hospitals: Where OSA-Shapley Loses Its Advantage

To study small silos without simulating any, we value the ten smallest PCa-19 hospitals (16–266 patients; exact non-private retraining Shapley as reference, 20 held-out splits). Without DP, OSA-Shapley is again the most accurate valuation ($\rho = 0.95$; OSA$-$MR $= +0.10$ $[+0.05, +0.14]$). Under DP its advantage disappears: at $\varepsilon = 2$ OSA and MR agree with the ranking equally well (0.79 and 0.78; difference $+0.00$ $[-0.06, +0.06]$), and at $\varepsilon = 1$ MR is slightly, not significantly, ahead (-0.06 [-0.15, +0.03]). Budget-matched retraining remains far worse ($\rho = 0.45$ at $\varepsilon = 2$). This is the regime that Proposition 3 predicts: the release of a hospital with a few dozen patients carries noise of order σ/n_k, whereas reconstruction averages over the federation in every round. With hospitals this small, hospital size is itself a strong proxy (Section VI-E), and no privacy-honest valuation improves on it at $\varepsilon \leq 1$.

I. A Second Real Federation: Heart Disease

Table IV repeats the comparison on the four hospitals of the UCI Heart Disease collection, with OSA-Shapley's PCa-19 hyper-parameters unchanged. Without DP the picture is the same as before: OSA-Shapley recovers the exact ranking best ($\rho = 0.92$, against 0.70 for MR and 0.76 for the single-model score; OSA$-$MR $= +0.22$ $[+0.05, +0.39]$). Under DP it is not. These hospitals have 98–242 training patients for 21 model parameters, and their contributions are nearly equal (Shapley values 0.085–0.101 AUC), so the noise of a single release dominates. At $\varepsilon = 5$ OSA-Shapley and MR are close (0.51 and 0.64); at $\varepsilon = 2$ MR is better (0.45 vs 0.15; OSA$-$MR $= -0.30$ $[-0.59, -0.01]$); and at $\varepsilon = 1$ no valuation agrees with the ranking beyond chance. Retraining does not help: with four hospitals its composed cost is modest (a stated $\varepsilon = 2$ really costs 6.1), yet per-run and budget-matched retraining reach only -0.09 and -0.18. Hospital size is anti-correlated with contribution ($\rho = -0.38$): the smallest hospital, Zurich/Basel, contributes most. Released models still detect bad data: OSA-Shapley and the single-model score rank Budapest with random labels last in 100% and 100% of splits at $\varepsilon = 2$, hospital size in 0%.

VII. A Decision Rule, Tested in Advance

A. The Index and the Original Rule

Supplementary Table S13 puts the four settings side by side. By Proposition 3 the per-coordinate noise of a release scales as $1/n_k$, so its norm over p parameters scales as $\sqrt{p}/n_k$; we therefore index a setting by ν, the median number of training patients per hospital divided by $\sqrt{p}$. In these settings OSA-Shapley won at $\nu = 114$ (ten largest PCa-19 hospitals) and did not at $\nu \leq 45$ (ten smallest: 45; heart disease: 43; MLP: 26), which suggested a switch-over between $\nu \approx 50$ and 100.

B. A Pre-Registered Test

A threshold read off the same results it explains is not evidence, so we tested it on new conditions. Before running anything, we wrote down the design and six hypotheses and fixed them with a cryptographic hash (Supplementary Section S4); the protocol was registered internally, not on a public registry, and written and run on the same day. Within the ten largest PCa-19 hospitals we lowered ν in two ways: arm A keeps 60% to 15% of every hospital's training patients, and arm B adds 21 to 100 uninformative features; in each condition the reference is the exact non-private retraining Shapley ranking of that condition (Fig. 2, Table S14).

The direction of the rule held, but its threshold did not. As predicted, OSA-Shapley beat MR at $\nu \geq 100$ (H1), its advantage shrank with ν (H3; Spearman correlation 0.57), both arms agreed in sign at matched ν (H5), and stronger privacy moved the switch-over to higher ν (H6). Contrary to H2 and H4, OSA-Shapley kept a significant advantage down to $\nu \approx 45$ (OSA$-$MR $= +0.18$ $[+0.08, +0.29]$ at $\varepsilon = 2$), tied with MR at $\nu \approx$ 30–57 when parameters were added, and fell behind only at $\nu \approx 17$ (OSA$-$MR $= -0.09$ $[-0.20, +0.03]$ at $\varepsilon = 2$, -0.18 $[-0.32, -0.03]$ at $\varepsilon = 1$). The mean difference changed

TABLE III

IS OSA-SHAPLEY WORTH ITS COST? AGREEMENT (SPEARMAN ρ) OF CHEAP CONTRIBUTION PROXIES AND OF PRIVACY-HONEST VALUATIONS WITH THE EXACT NON-PRIVATE RETRAINING SHAPLEY RANKING, ON REAL PCA-19 HOSPITALS (20 HELD-OUT SPLITS). *What is shared*: WHAT EACH HOSPITAL MUST REVEAL. PROXIES THAT USE A MODEL USE THE SAME DP RELEASES AS OSA-SHAPLEY AND THEREFORE COST THE SAME ε; SIZE, COUNTS AND STATISTICS ARE SHARED WITHOUT DP. *Useless*: SHARE OF SPLITS IN WHICH A HOSPITAL WHOSE TRAINING LABELS WERE REPLACED BY RANDOM LABELS (ES6) IS RANKED LAST AT $\varepsilon = 2$; A TIE FOR LAST PLACE DOES NOT COUNT.

		Ten largest hospitals				Ten smallest hospitals		
Contribution score	What is shared	no DP	$\varepsilon = 2$	$\varepsilon = 1$	Useless	no DP	$\varepsilon = 2$	$\varepsilon = 1$
Hospital size n_k	size (public)	0.08	0.08	0.08	0%	0.74	0.74	0.74
Number of csPCa cases	one count	0.02	0.02	0.02	0%	0.50	0.50	0.50
Similarity of summary statistics	feature means, prevalence	−0.18	−0.18	−0.18	0%	0.72	0.72	0.72
Self-reported local AUC	one AUC (unverifiable)	0.36	0.25	0.26	100%	0.40	0.54	0.56
Single-model AUC on reference set	one DP model (ε)	0.93	0.85	0.71	100%	0.94	0.79	0.72
Leave-one-out of averaged models	one DP model (ε)	0.85	0.77	0.74	0%	0.59	0.34	0.34
OR reconstruction	DP-FedAvg run (ε)	0.42	0.51	0.50	100%	0.79	0.74	0.59
MR reconstruction (GTG bound)	DP-FedAvg run (ε)	0.74	0.75	0.65	70%	0.86	0.78	0.74
Retraining, budget-matched	2^{K-1} runs (ε)	–	0.18	0.10	60%	–	0.45	0.29
OSA-Shapley (ours)	one DP model (ε)	**0.96**	**0.89**	**0.80**	100%	**0.95**	**0.79**	**0.68**

TABLE IV

SECOND REAL FEDERATION: THE FOUR HOSPITALS OF THE UCI HEART DISEASE COLLECTION (CLEVELAND (303), BUDAPEST (294), ZURICH/BASEL (123), LONG BEACH VA (200) PATIENTS), CORONARY ARTERY DISEASE, 20 HELD-OUT SPLITS. SPEARMAN ρ WITH THE EXACT NON-PRIVATE RETRAINING SHAPLEY RANKING; WITH FOUR HOSPITALS ρ MOVES IN STEPS OF 0.2. PER-RUN RETRAINING REALLY COSTS $\varepsilon \approx 16$, 6.1 AND 2.9 AT A STATED 5, 2 AND 1. *Useless*: SHARE OF SPLITS IN WHICH BUDAPEST, GIVEN RANDOM TRAINING LABELS, IS RANKED LAST AT $\varepsilon = 2$.

Contribution score	no DP	$\varepsilon = 5$	$\varepsilon = 2$	$\varepsilon = 1$	Useless
Hospital size n_k	−0.38	−0.38	−0.38	−0.38	0%
Similarity of summary statistics	−0.38	−0.38	−0.38	−0.38	0%
Self-reported local AUC	−0.18	−0.17	−0.31	−0.30	70%
Single-model AUC on reference set	0.76	0.42	−0.04	−0.34	100%
Leave-one-out of averaged models	0.82	0.69	0.42	0.18	85%
OR reconstruction	0.23	0.29	0.18	0.04	–
MR reconstruction (GTG bound)	0.70	0.64	0.45	−0.35	–
Retraining, per run	–	0.34	−0.09	−0.27	–
Retraining, budget-matched	–	−0.02	−0.18	−0.15	–
OSA-Shapley (ours)	**0.92**	**0.51**	**0.15**	**−0.18**	100%

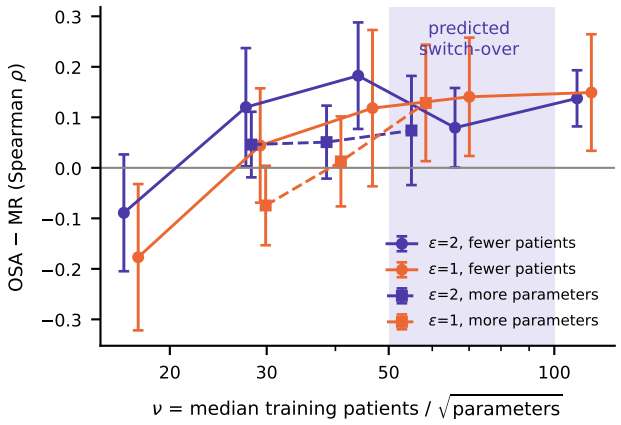


Fig. 2. Pre-registered test: OSA−MR (paired difference in Spearman ρ with the exact reference, 95% CI, 20 splits) as ν is lowered within the ten largest PCa-19 hospitals, by removing patients (solid) or adding uninformative parameters (dashed). Shaded: the switch-over predicted before the test. Points at the same ν are offset slightly for legibility.

sign at $\nu \approx 21$ for $\varepsilon = 2$ and between 26 and 38 for $\varepsilon = 1$, not between 50 and 100. Two of the six hypotheses therefore failed, and we revise the rule accordingly.

C. The Revised Rule

1) *Count the cost.* Report the composed ε of any valuation (Proposition 2) and do not retrain coalitions at a per-run budget; retraining was not accurate in any setting.
2) *Without DP on the releases* (for example, when model sharing is covered by a data-use agreement), use OSA-Shapley, the most accurate valuation in all four settings (ρ 0.91–0.96).
3) *Under DP with a convex model*, OSA-Shapley is at least as accurate as MR down to $\nu \approx 30$ at $\varepsilon = 2$ and $\nu \approx 40$ at $\varepsilon = 1$; below that, use MR, which adds no privacy cost when the federation already trains with DP-FedAvg.
4) *With a non-convex model*, use MR.
5) *Check hospital size before using it.* Size matched contribution only where hospitals differed mainly in volume (0.74 for the ten smallest PCa-19 hospitals) and failed elsewhere (0.08 and −0.38). Neither size nor summary statistics can detect bad data; the score of a single released model can.
6) *Compute both OSA-Shapley and MR when in doubt.* ν does not explain every case: the heart-disease federation ($\nu = 43$) favoured MR although the sweep says OSA should be competitive there. Its four hospitals make ρ coarse, and site-specific missing data make some features constant within a hospital, so single-hospital models are poorly determined; both explanations are untested. Disagreement between the two valuations is a sign that the ranking is not resolved at the chosen budget, and with small ν at $\varepsilon \leq 1$ coarse credit (equal shares plus screening for bad data) is the honest fallback.

VIII. DISCUSSION

The main lesson is about accounting. Studies of contribution valuation under DP routinely report a per-run privacy budget, but a hospital whose data enter 2^{K-1} coalition runs pays for all of them. On ten real hospitals that turns a stated $\varepsilon = 1$ into $\varepsilon \approx$

33, and when the budget is held honestly the ranking collapses. Any payment, reputation or cost-sharing scheme built on such valuations either over-spends privacy or rewards the wrong hospitals. The remedy is to restrict valuation to post-processing of releases whose cost is already counted.

Is it practical? OSA-Shapley asks each hospital for one DP-trained model, here seven numbers, which is the size of a single update in the 20-round DP-FedAvg run that the federation performs anyway; the 2^K coalition evaluations happen on the server and take about a minute for 19 hospitals. The cheaper alternative of sharing sizes, case counts or summary statistics is unreliable: these proxies were essentially uninformative for the large prostate-cancer hospitals ($\rho \leq 0.08$), anti-correlated with contribution in the heart-disease federation ($\rho = -0.38$), and blind to a hospital with useless data, because what a hospital adds depends on its case mix relative to the others, not on its size. Once a hospital has released a model, a single-model score is a reasonable approximation, which OSA-Shapley improves on at no extra privacy cost ($+0.10$ $[+0.05, +0.14]$ at $\varepsilon = 1$) while yielding proper Shapley values.

Within the privacy-honest class no single method is best, and Section VII turns our evidence into a rule. Its mechanism is simple: a valuation built on single releases is as good as those releases, whose noise grows with the number of parameters and shrinks with the number of patients, whereas reconstruction trades some bias (reconstructed models carry information from hospitals outside the coalition) for averaging over the whole federation in every round. Which effect dominates depends mainly on ν, although ν did not explain the heart-disease result (Section VII).

We avoid general statements about ε: the same budget barely changes the seven-parameter prostate calculator but degrades the heart-disease model and the MLP considerably, so results must be read per model and data size. Calibration, finally, is a local matter: the federated calculator transports in discrimination but not in calibration, and recalibrating the intercept on a hospital's own patients, already common practice before a risk calculator is used, removes most of the miscalibration (Supplementary Material).

A. Threats to Validity

Internal. OSA-Shapley's local steps were tuned on four splits and evaluated on twenty disjoint ones. The reference for valuation accuracy is exact non-private retraining Shapley over the ten largest and the ten smallest PCa-19 hospitals and all four heart-disease hospitals; for all 19 hospitals no exact retraining reference exists, so we report agreement with the non-private OSA ranking. PCa-19 data were harmonised by the base paper's authors from published supplementary files, and two hospitals have fewer than 50 patients.

External. All evaluations are retrospective and use two federations and two clinical tasks; with four hospitals, Spearman ρ moves in steps of 0.2, so the heart-disease estimates are coarse. One-shot averaging approximates the federated solution well for regularised logistic regression; for a federated MLP it is accurate without DP but loses to reconstruction under DP, and deeper networks are untested. Shapley values depend on the reference population on which coalitions are scored, and we have not tested a reference set from outside the federation.

Privacy. The guarantee is per patient and treats hospital sizes and shared preprocessing statistics as public. OSA-Shapley's guarantee covers the valuation release; if the training run is also released, the two budgets compose. Like every Shapley method, it scores coalitions on a reference set held by the server, whose patients are not protected unless it is public or itself privatised. The main experiments use full-batch updates with conservative accounting.

Statistical. Comparisons are paired over 20 splits that share the same patients, so CIs describe variability across splits rather than across federations.

B. Data, Code, Ethics and Conflicts

Both data sets are public and were used as released: PCa-19 from the base study's repository [3] and the UCI Heart Disease collection (doi:10.24432/C52P4X) [38]. They are de-identified, and no new patient data were collected, so no ethics approval was required. The authors declare no conflicts of interest and received no external funding. All results, tables and figures are regenerated from code: `python run_pca19.py` recomputes every experiment, `python -m fl_study.validate` runs 41 automated checks (data integrity, agreement with the base paper's code, the accountants, Propositions 2 and 3, and absence of leakage), and `python Paper/generate_paper_assets.py` writes every number in this manuscript. Code: https://github.com/JeyanthPonnaluri/Mini_Project_RVR.

IX. Conclusion

Valuing hospitals by retraining coalitions under differential privacy spends the privacy budget hundreds of times over, and at an honest budget it loses the contribution ranking; Monte-Carlo sampling reduces this cost only for large federations and never removes it. Valuation should therefore post-process releases whose cost is already counted. Among such valuations none wins everywhere. On two real federations, OSA-Shapley, which averages one DP-trained model per hospital, is the most accurate without DP and, under DP, when hospitals hold many patients per model parameter; reconstruction from the training run is better for a neural network and in the heart-disease federation, an exception the rule does not yet explain; hospital size and summary statistics are unreliable and cannot detect bad data. Patients per hospital relative to the square root of the model's parameters predicts the choice in direction; a pre-registered test moved the switch-over to about 30 at $\varepsilon = 2$ and 40 at $\varepsilon = 1$, and a federation near it should compute both valuations and report the composed privacy cost of whatever it runs. Explaining the heart-disease exception, OSA variants for neural networks, and prospective incentive schemes are the next steps.

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